GITAM Deemed to be University

Electronics and Communication Engineering | Course Code: EC305

Electromagnetic Waves & Transmission Lines - Laboratory Manual & Practic

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Syllabus Focus: EC305 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Maxwell Equations & Electromagnetic Statics

" Syllabus Scope: Gauss, Faraday, and Ampere laws in differential and integral forms, Displacement curren

Unit 2: Uniform Plane Wave Propagation

" Syllabus Scope: Helmholtz wave equation, Propagation constant, Attenuation and phase constants, Intrins

EC305: Electromagnetic Waves & Transmission Lines - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Reflection & Refraction at Media Boundaries

" Normal and oblique incidence, Snell law, Fresnel reflection and transmission coefficients, Brewster angle, T

Unit 4: Transmission Line Equations

" Transmission line differential equations, Characteristic impedance Z0, Propagation constant, Reflection coef

Unit 5: Smith Chart & Impedance Matching

" Smith chart coordinate geometry, Input impedance computation, Quarter-wave transformers, Single and double s

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Measurement of Standing Wave Pattern, VSWR, and Wavelength in a Waveguide Test Bench

Lab 2: Impedance Matching and Load Reflection Measurement using Smith Chart Simulation Tool

Lab 3: Measurement of Radiation Pattern and Directivity of a Half-Wave Dipole Antenna

Lab 4: Attenuation and Skin Depth Characterization in Coaxial and Microstrip Transmission Lines

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Matthew N.O. Sadiku - Elements of Electromagnetics Oxford University Press 7th Ed

* E.C. Jordan, K.G. Balmain - Electromagnetic Waves and Radiating Systems Prentice Hall

Authorized NoteVault Educational Document " Verified for Registered University Students